

$$\int_{-\infty}^{\infty} x^{2n} e^{-\alpha x^2} dx = \frac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{2^{n+1} \alpha^n} \left(\frac{\pi}{\alpha}\right)^{1/2}$$

$$\int_0^{\infty} x^{2n+1} e^{-\alpha x^2} dx = \frac{n!}{2 \alpha^{n+1}}$$

$$\int_0^{\infty} x^n e^{-\alpha x} dx = \frac{n!}{\alpha^{n+1}}$$

8.1 Introduction

Physics Problem: ODE

1) Satisfy BCs } Hilbert space + Eigenvalue Problem
 2) Parameter

Basic Example 8.1.1

$$\frac{d^2 \psi}{dx^2} + k^2 \psi = 0$$

$$\mathcal{L} \psi = k^2 \psi$$

$$\psi(0) = \psi(L) = 0$$

$$\mathcal{L} = -\frac{d^2}{dx^2}$$

Matrix Method

1) basis 2) inner product

3) L & ψ in basis 4) Matrix eqn

~~X~~ not optimal

ODE solns



$$\frac{d^2\psi}{dx^2} + k^2\psi = 0 \Rightarrow \text{general soln. } \psi(x) = A \sin kx + B \cos kx$$

$$\psi(0) = 0 \Rightarrow B = 0$$

$$\psi(l) = 0 \Rightarrow \sin kl = 0 \Rightarrow kl = n\pi$$

8.2 Hermitian Operator

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Sturm-Liouville Problem

$$\mathcal{L}\psi(x) = \lambda \psi(x)$$

2nd order diff op

$$\mathcal{L}(x) = p_0(x) \frac{d^2}{dx^2} + p_1(x) \frac{d}{dx} + p_2(x)$$

Self-adjoint DE

Condition $\mathcal{L}$ is SA if $p_0'(x) = p_1(x)$

$$\mathcal{L}(x) = \frac{d}{dx} \left[p_0 \frac{d}{dx} \right] + p_2(x)$$

$$\mathcal{L}u = [p_0 u']' + p_2 u$$

Proof

$$\langle v | Lu \rangle = \int_a^b v^* [(p_1 u')' + p_2 u] dx$$

 $p_0(x)$ is real

$$= \int_a^b [v^* (p_1 u')' + v^* p_2 u] dx$$

$$= [v^* p_0 u']_a^b + \int_a^b [- (v^*)' p_0 u' + v^* p_2 u] dx$$

$$= [v^* p_1 u' - (v^*)' p_1 u]_a^b + \int_a^b [(p_0 (v^*)')' u + v^* p_2 u] dx$$

$$\text{BC} \quad = \cancel{[v^* p_1 u' - (v^*)' p_1 u]_a^b} + \int_a^b (Lv)^* u dx = \langle Lv | u \rangle$$

BCs:

1) Dirichlet BCs: u, v vanish at a, b

2) Neumann BCs: u', v' vanish at a, b

3) $v^* \frac{d}{dx} u' = v^* p \cdot u'$ at a, b
for all u & v

$$\langle v | Lu \rangle = \left[v^* u' - (v^*)' p u \right]_a^b + \int_a^b (Lv)^* u dx = \langle Lv | u \rangle$$

for eigenfunctions

$$(\lambda_u - \lambda_v) \int_a^b v^* u dx = \left[v^* u' - (v^*)' p u \right]_a^b$$

$$\langle f | g \rangle = \int_a^b f^* g dx$$

Self-adjoint SL problem

$$L(x) = p_2(x) \frac{d^2}{dx^2} + p_1(x) \frac{d}{dx} + p_0(x)$$

$$p_0'(x) = p_1(x)$$

$$\langle f|g \rangle = \int f^* g dx$$

$$(\lambda_u - \lambda_v) \int_a^b v^* u dx = \left[\cancel{v^* p_2 u'} - (v^*)' p_2 u \right]_a^b$$

BCs

Making an ODE Self-adjoint

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Non self-adjoint $\Rightarrow$ multiply by $w(x) \equiv$ weight function

$$\text{Choose } w(x) = \frac{1}{p_0(x)} \exp\left(\int \frac{p_1(x)}{p_0(x)} dx\right)$$

$$w(x) \mathcal{L} \psi(x) = w(x) \lambda \psi(x)$$

$$w(x) \mathcal{L}(x) = \bar{p}_0 \frac{d^2}{dx^2} + \bar{p}_1 \frac{d}{dx} + w(x) p_2(x)$$

$$\bar{p}_0 = e^{\int \frac{p_1}{p_0} dx} \quad \bar{p}_1 = \frac{p_1}{p_0} e^{\int \frac{p_1}{p_0} dx} \Rightarrow \bar{p}_0' = \bar{p}_1$$

$$\Rightarrow \int_a^b v^* w \mathcal{L} u \, dx = \left[v^* \bar{p}_1 u' - (\overset{\circ}{v^*})' \bar{p}_1 u \right]_a^b + \int_a^b w (\mathcal{L} v)^* u \, dx$$

$$\langle v | \mathcal{L} u \rangle = \langle \mathcal{L} v | u \rangle$$

Assuming $\langle v | u \rangle = \int_a^b v^* u w \, dx$

Eigenfunctions $(\lambda_u - \lambda_v) \int_a^b v^* u w \, dx = \left[w \bar{p}_1 (v^* u' - (\overset{\circ}{v^*})' u) \right]_a^b$

$$L(x) = p_0 \frac{d^2}{dx^2} + p_1 \frac{d}{dx} + p_2$$

$$p_0' \neq p_1 \Rightarrow \text{multiply by } w(x) = \frac{1}{p_0(x)} \exp\left(\int \frac{p_1}{p_0} dx\right)$$

$$\langle f | g \rangle = \int_a^b f^*(x) g(x) w(x) dx$$

$$\underline{\text{Eigenfunctions}} \quad (\lambda_u - \lambda_v) \int_a^b v^* u w dx = \left[w p_0 (v^* u' - (v^*)' u) \right]_a^b$$

Ex 8.2.1

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Laguerre functions

$$\mathcal{L} = x \frac{d^2}{dx^2} + (1-x) \frac{d}{dx}$$

a) ψ nonsingular on $0 \leq x < \infty$

b) $\lim_{x \rightarrow \infty} \psi(x) = 0$

$\mathcal{L}$ is not SA $p_0 = x$, $p_1 = 1-x \Rightarrow w = \frac{1}{x} e^{-x}$

$$w(x) = \frac{1}{x} e^{\ln x - x} = e^{-x}$$

$$\left. \begin{array}{l} \text{BC}_s \\ [x e^{-x} (v^* u' - (v^*)' u)]_0^\infty = 0 \end{array} \right\} \langle v | u \rangle = \int_0^\infty v^* u e^{-x} dx$$

8.3 ODE Eigenvalue Problem

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Ex 8.3.1 Legendre Eqn

$$L y(x) = -(1-x^2) y''(x) + 2x y'(x) = \lambda y(x)$$

$$-1 \leq x \leq 1$$

Solns that are nonsingular

L is Hermitian

$$* p_0' = p_1$$

$$* p_2(\pm 1) = 0$$

Ex 8.3.1 Legendre Eqn

$$L y(x) = -(1-x^2) y''(x) + 2x y'(x) = \lambda y(x)$$

$$-1 \leq x \leq 1$$

Method of Frobenius

$$y = \sum_{j=0}^{\infty} a_j x^{s+j} \quad y' = \sum_{j=0}^{\infty} a_j (s+j) x^{s+j-1} \quad y'' = \sum_{j=0}^{\infty} a_j (s+j)(s+j-1) x^{s+j-2}$$

$$\sum_{j=0}^{\infty} a_j x^{s+j} [(x^2-1)(s+j)(s+j-1) x^{-2} + 2x(s+j) x^{-1} - \lambda] = 0$$

$$\sum_{j=0}^{\infty} a_j x^{s+j} \left\{ -(s+j)(s+j-1) x^{-2} + \left[(s+j)(s+j-1) + 2(s+j) - \lambda \right] \right\} = 0$$

lowest power ($j=0$) x^{s-2}

$$s(s-1) = 0$$

$$s = 0, 1$$

Indicial eqn.

$$\underline{s=0} \quad -\sum_{j=0}^{\infty} a_j x^{j-2} j(j-1) + \sum_{j=0}^{\infty} a_j x^j [j(j+1) - \lambda] = 0$$

$$j=0 \quad \checkmark$$

$$a_0$$

$$j=1$$

$$a_1 = 0$$

$$k = j - 2 \quad j = k + 2$$

$$-\sum_{k=0}^{\infty} a_{k+2} x^k (k+1)(k+2) + \sum_{j=0}^{\infty} a_j x^j [j(j+1) - \lambda] = 0$$

$$-\sum_{j=0}^{\infty} a_{j+2} x^j (j+1)(j+2) + \sum_{j=0}^{\infty} a_j x^j [j(j+1) - \lambda] = 0$$

$$\sum_{j=0}^{\infty} x^j [- (j+1)(j+2) a_{j+2} + \{ j(j+1) - \lambda \} a_j] = 0$$

$$a_{j+2} = \frac{j(j+1) - \lambda}{(j+1)(j+2)} a_j$$

"Recurrence relation"

BC (± 1) $\Rightarrow$ Series soln. at $\sum_{j=0}^{\infty} a_j$ diverges

we have to truncate series

Terminate at $a_l \Rightarrow a_{l+2} = 0$

$$\lambda = l(l+1)$$

$\Rightarrow$ Eigenvalues $\lambda = l(l+1)$ even l

Eigenfunctions P_l Legendre Poly.

$$\underline{S=1}$$

$$- \sum_{j=0}^{\infty} a_j x^{j+1} j(j+1) + \sum_{j=0}^{\infty} a_j x^{j+1} [j(j+1) + 2(j+1) - \lambda] = 0$$

$$j=0$$

$$j=1$$

$$a_1 = 0$$

$$k = j - 2$$

$$j = k + 2$$

$$\sum_{j=0}^{\infty} x^{j+1} [-a_{j+2} (j+2)(j+1) + \{ (j+1)(j+2) - \lambda \} a_j] = 0$$

$$a_{j+2} = \frac{(j+1)(j+2) - \lambda}{(j+2)(j+1)} a_j$$

$$\text{Terminate } a_l \Rightarrow \lambda = (l+1)(l+2) \quad l \text{ even } \left(\begin{matrix} x^{l+1} \\ \text{odd} \end{matrix} \right)$$

Replace $(l+1)$ with l

Eigenvalues $\lambda = l(l+1)$

odd l

Eigenfunction P_l

Example 8.3.2

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Hermite Eqn

$$\mathcal{L}y = -y'' + 2xy' = \lambda y \quad -\infty < x < \infty$$

Hermitian? No $\Rightarrow w(x) = e^{-x^2}$

Scalar Product $\langle f | g \rangle = \int_{-\infty}^{\infty} f^*(x) g(x) \underline{\underline{e^{-x^2}}} dx$

$$y_n \Leftrightarrow \langle y_n | y_n \rangle < \infty$$

Indicial eqn

$$s(s-1) = 0$$

$s = 0$

$$a_{j+2} = \frac{2j-1}{(j+1)(j+2)} a_j$$

$$\sum_{j=0}^{\infty} a_j x^j \sim e^{x^2}$$

Terminat at $a_n \Rightarrow \lambda = 2n$ even $p=1$.

$s = 1$

odd $p=1$.

Hermite Polynomials